INCH-POUND
MIL-DTL-24643/24C
1 October 2009
SUPERSEDING
MIL-DTL-24643/24B
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 °C TO +150 °C, TYPES LSTCJX, LSTCKX, LSTCTX, LSTCJXN, LSTCKXN, AND LSTCTXN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction,
Types LSTCJX, LSTCKX and LSTCTX (Watertight with Circuit Integrity)
Types LSTCJXN, LSTCKXN and LSTCTXN (Non-Watertight with Circuit Integrity)

First	-	Stranded conductor, uncoated (see table I for size and stranding).
Second	-	Extruded silicone rubber insulation (see table I for dimensions).
Third	-	Glass braid (see table I for dimensions).
Fourth	-	Transparent braid covering.
Fifth	-	Two conductors cabled together to form a pair with a lay not greater than 2.25 inches. For LSTCJX and LSTCJXN, one iron conductor and one constantan conductor. For LSTCKX and LSTCKXN, one chromel and one alumel conductor. For LSTCTX and LSTCTXN, one copper conductor and one constantan conductor. Types J, K, and T thermocouple conductors shall be in accordance with ANSI MC96.1. Color code applied by Method 4, see table II for color code sequence.
Sixth	-	The specified number of pairs (see table I) shall be cabled together with a lay not greater than 10 inches. Fillers may be used as necessary to form a firm well-rounded assembly. Cabling sequence shall be consecutive, starting with pair 1 from the center outward.
Seventh	-	A binder applied helically with overlap.
Eighth	-	Silicone rubber jacket colored orange or red (see table I for thickness).
Ninth	-	Braided metal armor.

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TABLE I. Details.

Military part no. M24643/24	Type and size	Conductors				Overall cable	
		No. of pairs	No. of strands and strand diameter (inch)	Diameter over insulation nominal (inch)	Diameter over braid covering max. (inch)	Jacket thickness min. avg. (inch)	Diameter max. (inch)
-01AN	LSTCJX-3	3	7/0.0201	0.100	0.125	0.050	0.742
-02AN	LSTCJX-7	7	7/0.0201	0.100	0.125	0.062	0.983
-03AN	LSTCJX-12	12	7/0.0201	0.100	0.125	0.062	1.269
-04AN	LSTCKX-1	1	7/0.0201	0.100	0.125	0.038	0.456
-05AN	LSTCKX-3	3	7/0.0201	0.100	0.125	0.050	0.742
-06AN	LSTCKX-7	7	7/0.0201	0.100	0.125	0.062	0.983
-07AN	LSTCKX-12	12	7/0.0201	0.100	0.125	0.062	1.269
-08AN	LSTCTX-1	1	7/0.0113	0.065	0.085	0.038	0.350
-09AN	LSTCTX-3	3	7/0.0113	0.065	0.085	0.038	0.552
-10AN	LSTCTX-7	7	7/0.0113	0.065	0.085	0.050	0.731
-11AN	LSTCTX-12	12	7/0.0113	0.065	0.085	0.063	0.964
-12AN	LSTCJXN-3	3	7/0.0201	0.100	0.125	0.050	0.742
-13AN	LSTCJXN-7	7	7/0.0201	0.100	0.125	0.062	0.983
-14AN	LSTCJXN-12	12	7/0.0201	0.100	0.125	0.062	1.269
-15AN	LSTCKXN-1	1	7/0.0201	0.100	0.125	0.038	0.456
-16AN	LSTCKXN-3	3	7/0.0201	0.100	0.125	0.050	0.742
-17AN	LSTCKXN-7	7	7/0.0201	0.100	0.125	0.062	0.983
-18AN	LSTCKXN-12	12	7/0.0201	0.100	0.125	0.062	1.269
-19AN	LSTCTXN-1	1	7/0.0113	0.065	0.085	0.038	0.350
-20AN	LSTCTXN-3	3	7/0.0113	0.065	0.085	0.038	0.552
-21AN	LSTCTXN-7	7	7/0.0113	0.065	0.085	0.050	0.731
-22AN	LSTCTXN-12	12	7/0.0113	0.065	0.085	0.063	0.964

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TABLE II. Color identification code.

Pair number	Constantan or alumel	Iron, copper, or chromel leg	
		Base color	Tracer color
1	Red	White	-
2	Red	Black	-
3	Red	Green	-
4	Red	Orange	-
5	Red	Blue	-
6	Red	White	Black
7	Red	White	Green
8	Red	White	Orange
9	Red	White	Blue
10	Red	White	Red
11	Red	Black	White
12	Red	Green	White

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic Electricals:

Voltage withstand (volts, root mean square, min.)

Conductor to conductor 2000

Insulation resistance (megohms/1000 feet, min.)

Conductor to conductor 500

Conductor continuity No failure

ANSI MC96.1 calibration limits of error:

Types LSTCJX and LSTCJXN ±2.2 °C

Types LSTCKX and LSTCKXN -32 to 530 °F ±4 °C

Types LSTCTX and LSTCTXN ±1 °C

Group A:

Visual and dimensional No failure

Watertightness – Types LSTCJX, LSTCKX, and LSTCTX only (see MIL-DTL-24643 for limits of water leakage) No failure

Group B:

Cold bending cable No failure

Thermoset proof test (percent, max.)

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Insulation	50
Jacket (when tested at 200 °C)	50
Physicals (unaged)	
Insulation (extruded)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	800
Elongation (percent, min.)	250
Durometer (jacket) - Type A (hardness, min.)	60
Group C:	
Physicals, cable aging (260±10 °C)	
Jacket (cable)	
Tensile strength (lb/in^2, min.)	600
Elongation (percent, min.)	150
Group D:	
Flame propagation (cable)	No failure
Qualification Inspection:	
Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:	
Gas flame (3 hours)	No failure
Aging and compatibility (cable) (125±5 °C)	No failure
Armor (conformance to material, construction and coverage)	No failure
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Smoke index (max.)	
Jacket	35
Fillers	45

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Insulation	35
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
All types and sizes	100

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2008-028)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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